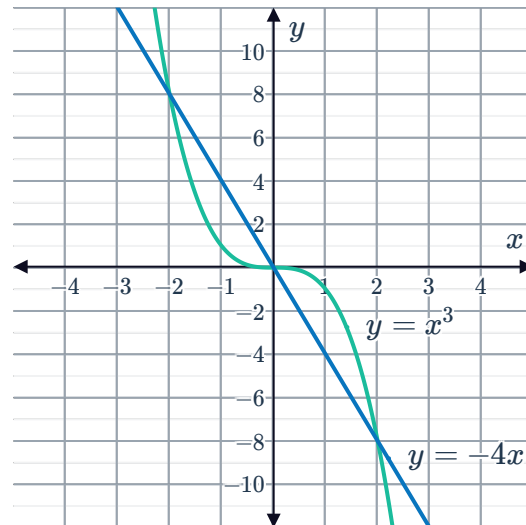


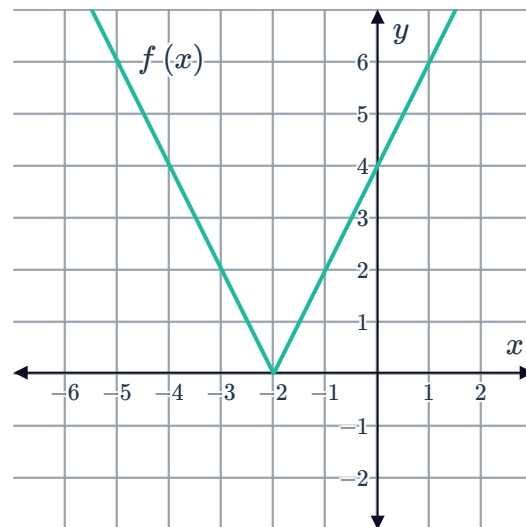
## Solve equations graphically



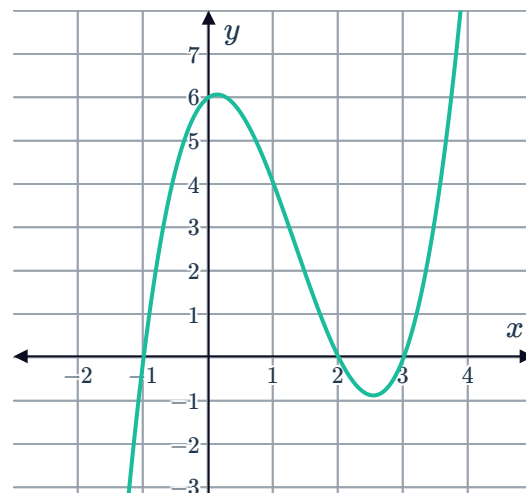
- 1 The functions  $y = -x^3$  and  $y = -4x$  are shown on the following graph:  
How many values of  $x$  satisfy the equation  $-x^3 = -4x$ ?



- 2 Consider the following graph of the function  $f(x) = |2x + 4|$ :
- Sketch the graph of  $g(x) = 2$  on the same coordinate plane as  $f(x)$ .
  - State the number of solutions for the equation  $|2x + 4| = 2$ .
  - Find the solutions to the equation  $|2x + 4| = 2$ .



- 3 Consider the graph of the cubic function  $y = (x + 1)(x - 2)(x - 3)$ :
- State the number of solutions to the equation  $(x + 1)(x - 2)(x - 3) = 0$ .
  - Determine the solutions to the equation  $(x + 1)(x - 2)(x - 3) = 0$ .
  - State the number of solutions to the equation  $(x + 1)(x - 2)(x - 3) = -2$ .
  - State the number of solutions to the equation  $(x + 1)(x - 2)(x - 3) = 6$ .





- 4 Consider the functions  $y = 9$  and  $y = x^2$ .
- a Sketch the graph of the functions on the same coordinate plane.
  - b State the coordinates of the points of intersection.
  - c Now, state the solutions to the equation  $x^2 = 9$ .
- 5 Consider the quadratic function  $y = x^2 + 4$  and the linear function  $x + y = 4$ .
- a Sketch the graph of the functions on the same coordinate plane.
  - b Now, state the number of solutions to the equation  $x^2 + 4 = -x + 4$ .
  - c Determine the coordinates of the points of intersection of the two functions.
- 6 Consider the quadratic function  $y = x^2 - 3$  and the linear function  $x + y = 3$ .
- a Sketch the graph of the functions on the same coordinate plane.
  - b Now, state the number of solutions to the equation  $x^2 - 3 = -x + 3$ .
  - c Determine the coordinates of the points of intersection of  $y = x^2 - 3$  and  $x + y = 3$ .
- 7 Consider the functions  $f(x) = -4x + x^2$  and  $g(x) = -12 + 6x - x^2$ .
- a Sketch the graph of  $f(x)$  and  $g(x)$  on the same coordinate plane.
  - b Determine the coordinates of the points of intersection of  $f(x)$  and  $g(x)$ .
  - c Now, determine the solutions to the equation  $-4x + x^2 = -12 + 6x - x^2$ .



## Solve equations using technology

8 Use technology to solve the following equations. Round your answer to two decimal places where necessary.

a  $x - 5 = \sqrt{x + 3}$

b  $\sqrt[3]{x + 13} = 2$

c  $x^{\frac{5}{3}} = 5 - 4x^2$

d  $\sqrt{x + 4} + \sqrt{4x + 3} = 3$

e  $\sqrt{3x - 2} = \sqrt{x} - \frac{1}{2}$

f  $3^x = \frac{1}{x}$

g  $2^{x+2} = 3^{x+3}$

h  $2x - 5 = |x - 4|$

i  $|x + 4| - 6 = -|x + 2| + 6$

j  $\frac{3x^3}{x + 2} = \frac{4x^2}{x - 5}$

k  $3x - 2 = 5x^3 + 2x^2 - x - 6$

l  $3x^2 - 2x - 40 = -x^2 + 2x + 8$

m  $2x^4 + 5x^3 + x = 3x^4 + 12x^3 - 81x^2 - 566x$

n  $2x^3 + 3x^2 + 4x + 5 = 7x^3 + 6x^2 + 5x + 4$

9 Solve  $3^{-x} = \sqrt{x + 4}$ . Round your answer to three decimal places.



## Applications

10 Ned is looking to find approximate solutions to the simultaneous equations:

$$f(x) = -2^x + 9$$
$$g(x) = x^3 + 3x + 2$$

- a Given that  $h(x) = g(x) - f(x)$ , write an expression for  $h(x)$  in simplified form.
- b Complete the following table of values for  $h(x)$ :

|        |    |    |   |   |   |
|--------|----|----|---|---|---|
| $t$    | -2 | -1 | 0 | 1 | 2 |
| $h(x)$ |    |    |   |   |   |

- c The value of  $x$  that satisfies  $f(x)$  and  $g(x)$  simultaneously lies in which interval?

11 The populations, in thousands, of two competing species of animals is given by  $f(t) = 215t + 3$  and  $g(t) = \frac{1}{t-3}$ , where  $f(t)$  and  $g(t)$  represent the respective populations  $t$  years from now. Jenny is looking to determine approximately when their populations will be equal.

- a She first equates the two functions and forms a new quadratic equation  $h(t)$ . Write an equation for  $h(t)$  in the form  $at^2 + bt + c$ , where  $a > 0$ .
- b Complete the table of values, rounding the answer to two decimal places if necessary:

|        |   |      |     |      |   |
|--------|---|------|-----|------|---|
| $t$    | 3 | 3.25 | 3.5 | 3.75 | 4 |
| $h(t)$ |   |      |     |      |   |

- c Using the table from part (b), state what value of  $t$  should Jenny use to give an approximation to the point of intersection of  $f(t)$  and  $g(t)$ .
- d The equation  $h(t) = 0$  has two solutions, but Jenny ignored the other solution. Explain why the other solution was ignored.